User Lands on Entropy (AWS Amplify URL)

Authentication Required?

- No → Continue as Guest (Anonymous Session)
- Yes → Sign In / Sign Up (AWS Cognito / Email)

Welcome Screen
- Tutorial Modal (First Visit)
- Example Commands List
- Safety Notice: 'All commands run in isolation'
- Button: 'Start Sandbox'

Backend Creates Session
1. Generate sessionID (UUID)
2. Spawn isolated Docker container
3. Establish WebSocket connection
4. Display terminal: 'root@ops-guru:~#'

Main Entropy Interface

```
┌──────────────────────────────┬──────────────────────────────┐
│ Terminal (Left 60%)          │ 3D Visualization (Right 40%) │
│ $ _                          │ [Green Cube Server]          │
│                              │                              │
├──────────────────────────────┴──────────────────────────────┤
│ AI Chat (Bottom Drawer)                                     │
│ 'Ready to explain your chaos!'                              │
└─────────────────────────────────────────────────────────────┘
```

User Action?

- Execute Command → User Types Command
  Example: 'rm -rf /usr'
  - Press Enter
  - WebSocket sends to backend
  → Backend Executes in Container
  1. Capture stdout/stderr
  2. Monitor syscalls (eBPF)
  3. Detect exit code != 0
  → Real-Time Visual Feedback
  - Terminal shows error output
  - 3D Cube turns RED
  - Shake animation plays
  - Error sound effect (optional)
  → AI Explanation (Auto-Trigger)
  - Bedrock analyzes error
  - Stream response token-by-token
  - Display in chat drawer

  Example:
  'You tried to delete /usr, which contains
  critical system binaries. This caused a
  SegFault. Fix: Don't target system dirs.'
- Try Scenario → User Clicks 'Scenarios' Tab
  - Guided Challenges:
    - High CPU Attack
    - Fork Bomb
    - OOMKilled Simulation
    - Kernel Panic (kill -9 1)
  → Load Pre-configured Scenario
  - Backend injects test script
  - AI provides context:
  'This will simulate OOM. Watch the kernel kill your process.'
  → User Executes Scenario
  - Automated command sequence
  - Real-time metrics overlay:
  CPU: 98%, RAM: 100%
  - Visual: Cube explodes animation
- Save Progress → User Clicks 'Save Progress'
  - Button in top-right corner
  - Backend triggers snapshot
  → Snapshot Process
  1. docker commit container
  2. Compress to .tar.gz
  3. Upload to S3
  4. Show toast: 'Saved successfully!'
  → User Returns Later
  - Logs back in
  - Dashboard shows saved sessions:
    - 'Session from Feb 12, 2026'
    - 'Fork Bomb Practice'
  - Click 'Restore'
  → Restore Complete
  - Backend loads snapshot from S3
  - Spawn new container from image
  - User sees exact same state:
  $ ls → my-awesome-project/
- Ask AI → User Asks AI Directly
  - Types in chat: 'What happens if I delete /etc/passwd?'
  - No command execution
  → AI Proactive Response
  - Bedrock generates educational answer
  - Shows example output
  - Offers 'Try it safely' button
  → User Clicks 'Try It'
  - Backend auto-fills terminal
  - User reviews and hits Enter
  - Loop back to execution flow

Continue or Exit?

- Exit → User Exits
  - Close browser tab OR
  - Click 'End Session' button
  → Backend Cleanup
  1. Close WebSocket
  2. Stop container (5 sec grace)
  3. Delete container (unless saved)
  4. Free resources
  5. Log session metrics to CloudWatch
  → Session Ended
  - Show stats:
    - Commands Run: 23
    - Crashes: 5
    - Time Spent: 12 min
  - Prompt: 'Save session?'
- Continue → Continue Experimenting
  - Reset button clears terminal
  - Keeps same container